The empirical recurrence for OEIS A236981: recovering a conjecture that is not written down

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Abstract

OEIS A236981 records "Empirical recurrence of order 92", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 332$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 92, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 92 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236981 is "Number of (n+1)X(2+1) 0..3 arrays with the maximum plus the upper median plus the lower median of every 2X2 subblock equal.". It has offset 1 and begins

$$888, 3852, 16212, 85064, 429108, 2380742, 12891190, 73033832, \ldots$$

The entry states:

> Empirical recurrence of order 92 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 09:42:54, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 332$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 332$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 664$ exact terms of the model returns order 92 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{92} c_i\, a(n-i),$$

c_1	31	c_{24}	−23238288143942194428	c_{47}	−4039693537207701305073747736
c_2	−186	c_{25}	9136557320061519751	c_{48}	8515735695926166881820217024
c_3	−3913	c_{26}	246145251817292576068	c_{49}	7272416280947321005313163808
c_4	54279	c_{27}	−281041603779708264846	c_{50}	−21120117625375217048181173296
c_5	97300	c_{28}	−2130947151661767242569	c_{51}	−9739684518006479526371525760
c_6	−5120647	c_{29}	3958279856510572086709	c_{52}	43912672910843514368930021696
c_7	13735401	c_{30}	15041765917745942993960	c_{53}	7279560161017176356225917088
c_8	263592270	c_{31}	−40030891835606461751221	c_{54}	−76830988504213936405554236672
c_9	−1524591017	c_{32}	−85104302245954326355321	c_{55}	5366476796245827335850223872
c_{10}	−8004981090	c_{33}	319733558687353853518459	c_{56}	113138777083363374988933566976
c_{11}	81255805082	c_{34}	367034924260697029707753	c_{57}	−30343837214039640007733018240
c_{12}	112626032357	c_{35}	−2092319373802213634156309	c_{58}	−139862868567236377566680538368
c_{13}	−2859609099075	c_{36}	−1010208808033368496834684	c_{59}	62211796746971153964899470848
c_{14}	1877025621850	c_{37}	11414621376719816842120818	c_{60}	144403692810327776564123948544
c_{15}	72746381595025	c_{38}	−225324246787363815989592	c_{61}	−88455067939448010330088624640
c_{16}	−156908614837670	c_{39}	−52354205325998980879311104	c_{62}	−123513832806814318042971801600
c_{17}	−1382163309546444	c_{40}	22745883095275096263419518	c_{63}	96962562709997641994725099520
c_{18}	4999352606511663	c_{41}	202489005735033515509394964	c_{64}	86454194679885577608561119232
c_{19}	19536090164374090	c_{42}	−163542454902175877341339536	c_{65}	−84806059105771796954004944896
c_{20}	−108155384395072315	c_{43}	−659467456679162271748666536	c_{66}	−48578481148246134660311359488
c_{21}	−192280175423281728	c_{44}	776298800176918596925406308	c_{67}	59917381908149486410760544256
c_{22}	1778771192230039125	c_{45}	1796992498816110471556798536	c_{68}	21193213835479758023981293568
c_{23}	909931868317562755	c_{46}	−2851204264649842047545052536	c_{69}	−34288825857891865456391487488

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 92, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $92 + 4$ terms removed returns order 92 as well, so $\deg q_{\min} = 92 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $92+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 92 that this sequence satisfies — and that family turns out to have exactly one member.

References

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